Senjuti Dutta

Postdoctoral Fellow

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Experience

- **University of Tennessee, Knoxville**, Knoxville, TN, USA, 08/2026 -.
  Postdoctoral Fellow of Center for the Dynamics of Social Complexity.
- **University of Colorado Boulder**, Boulder, CO, USA, 09/2025 – 07/2026.
  Postdoctoral Researcher at SHINE Lab.
- **xAI**, Burlingame, CA, USA, 11/2024 – 09/2025.
  Human Data STEM Specialist.
- **Keywords Studios**, Bothell, WA, USA, 05/2024 – 07/2024.
  Human Data Manager Intern.
- **University of Tennessee Knoxville**, Knoxville, TN, USA, 08/2020 – 08/2024.
  Graduate Research Assistant.
- **Google Research**, Mountain view, CA, USA, 08/2022 – 5/2023.
  Research Intern.
- **University of Memphis**, Memphis, TN, USA, 08/2017 – 05/2020.
  Research Assistant.
- **Jadavpur University**, West Bengal, India, 11/2016 – 07/2017.
  Graduate Research Assistant.

Fellowships, Awards, and Honors

- **Graduate Student Senate Award for Excellence in Graduate Research**, University of Tennessee, Knoxville **April 2024**
  Awarded in recognition of outstanding graduate research contributions across disciplines.
- **Yates Dissertation Fellowship**, University of Tennessee, Knoxville **August 2023**
  Competitive university fellowship awarded to outstanding doctoral candidates in recognition of academic excellence and to support dissertation research and completion.

Education

- **Ph.D., Computer Science**
  Department of Electrical Engineering and Computer Science, University of Tennessee, 2020 – 2024
  Advisor: Dr. Scott Ruoti
  Thesis Committee: Dr. Scott Ruoti, Dr. Alex Williams, Dr. Anastasia Kuzminykh, Dr. Audris Mockus

- **M.S., Computer Science**
  Department of Computer Science, University of Memphis, 2017 – 2020
  Advisor: Dr. Kan Yang.

- **B.S., Computer Engineering**
  Deptartment of Computer Science, MSIT, 2016 – 2020

Peer-Reviewed Selected Conference Papers

[C.8] **Dutta, S.**, Ruoti, S. (2025). A Comparative Study of Device Usage and Preferences Across Crowdsourcing Platforms. (Acceptance Rate: 24%)

[C.7] **Dutta, S.**, Linder, R., Williams, A.C., Kuzminykh, A., Ruoti, S., 2024, October. Unveiling the Inter-Related Preferences of Crowdworkers: Implications for Personalized and Flexible Platform Design. InProceedings of the AAAI Conference on Human Computation and Crowdsourcing (HCOMP 2024), Vol.12, pp.55–64.

[C.6] Collins, K.M., Kim, N., Bitton, Y., Rieser, V., Omidshafiei, S., Hu, Y., Chen, S., **Dutta, S.**, Chang, M., Lee, K., Liang, Y., 2024, October. Beyond Thumbs Up/Down: Untangling Challenges of Fine-Grained Feedback for Text-to-Image Generation. InProceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES2024), Vol.7, pp.293–303. (Acceptance Rate: 38%)

[C.5] Lee, M., Gero, K., Chung, J., Shum, S., Raheja, V., Shen, H., Venugopalan, S., Wambsganss, T., Zhou, D., Alghamdi, E., August, T., Bhat, A., Choksi, M., **Dutta, S.**, and others, 2024, May. A Design Space for Intelligent and Interactive Writing Assistants. InProceedings of the 2024 CHI Conference on Human Factors in Computing Systems, pp.1–35. (Acceptance Rate: 25.3%)

[C.4] **Dutta, S.**, Linder, R., Lowe, D., Rosenbalm, R., Kuzminykh, A., Williams, A.C., 2022, April. Mobilizing Crowdwork: A Systematic Assessment of the Mobile Usability of HITs. InProceedings of the 2022 CHI Conference on Human Factors in Computing Systems, pp.1–20. (Acceptance Rate: 25.3%)

[C.3] Linder, R., Hunter, C., McLemore, J., **Dutta, S.**, Akbar, F., Grover, T., Breideband, T., Borghouts, J.W., Lu, Y., Mark, G., Henley, A.Z., 2022. Characterizing Work-Life for Information Work on Mars: A Design Fiction for the New Future of Work on Earth.Proceedings of the ACM on Human-Computer Interaction,6(GROUP), pp.1–27.

[C.2] Yang, K., **Dutta, S.**, 2021, June. Secure and Efficient Task Matching with Multi-Keyword in Multi-Requester and Multi-Worker Crowdsourcing. In2021 IEEE/ACM 29th International Symposium on Quality of Service (IWQoS), pp.1–6.

[C.1] **Dutta, S.**, Barik, M.S., Chowdhury, C., Gupta, D., 2018, January. Divya-Dristi: A Smartphone-Based Campus Navigation System for the Visually Impaired. In2018 Fifth International Conference on Emerging Applications of Information Technology (EAIT), pp.1–3.

Peer-Reviewed Workshop Papers

[C.5] Collins, K., Kim, N., Bitton, Y., Rieser, V., Omidshafiei, S., Hu, Y., Chen, S., **Dutta, S.**, Chang, M., Lee, K., Liang, Y., Evans, G., Singla, S., Li, G., Weller, A., He, J., Ramachandran, D., Dvijotham, K. Beyond Thumbs Up/Down: Untangling Challenges of Fine-Grained Feedback for Text-to-Image Generation `(ICML2024Workshop)` .

[C.4] **Dutta, S.**, Linder, R., Williams, A.C., Kuzminykh, A., Ruoti, S. Design Implications for Developing Flexible and Inclusive Crowdwork AI Systems (HCOMP2024 Workshop).

[C.3] **Dutta, S.**, R.Linder, D.Lowe, M.Rosenbalm, A.Kuzminykh, A.Williams. The Productivity Paradox: Understanding Tooling Biases in Crowdwork (CSCW'21 Workshop– Investigating and Mitigating Biases in Crowdsourced Data).

[C.2] **Dutta, S.**, R.Linder, A.Kuzminykh, A.Williams. Doing Work from Where You Are: AI HIT Management Dispatching Work Tailored to Context (Accepted at the CHI'22 Crowd Science Workshop).

[C.1] Chen, S., Morgan, C., Olsen, D., Manilow, E., Allado-McDowell, K., Nelson, M.J., Zhang, Q.R., **Dutta, S.**, Mathewson, K. and Mirowski, P.W., Towards an Authorial Leverage Evaluation Framework for Expressive Benefits of Deep Generative Models in Story Writing..

Extended Abstracts

[C.1] **Dutta, S.**, Doherty, E., Hirshfield, L. (2026, April). Mapping the Spectrum of Automatability: A Framework for LLM-Assisted Annotation of Team Communication and Metacognition. In Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (pp. 1-28). (Acceptance Rate: 20-25%)

Peer-Reviewed Posters

[C.1] **Dutta, S.**, Linder, R., Ruoti, S., Williams, A., Kuzminykh, A. (2022, September). Beyond a one-size-fits-all approach: Towards personalizing multi-device setups in crowdwork. In Adjunct Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing and the 2022 ACM International Symposium on Wearable Computers (pp. 30-31). (Acceptance Rate: 29-34%)

arXiv Papers

[C.3] Elgedawy, R., Sadik, J., **Dutta, S.**, Gautam, A., Georgiou, K., Gholamrezae, F., Ji, F., Lim, K., Liu, Q., Ruoti, S., 2024. Occasionally Secure: A Comparative Analysis of Code Generation Assistants. arXiv preprint arXiv:2402.00689.

[C.2] **Dutta, S.**, Chen, S., Mak, S., Ahmad, A., Collins, K., Butryna, A., Ramachandran, D., Dvijotham, K., Pavlick, E., Rajakumar, R., 2024. Understanding Subjectivity through the Lens of Motivational Context in Model-Generated Image Satisfaction. arXiv preprint arXiv:2403.05576.

[C.1] **Dutta, S.**, Mittal, S., Chen, S., Ramachandran, D., Rajakumar, R., Kivlichan, I., Mak, S., Butryna, A., Paritosh, P., 2023. Modeling Subjectivity (by Mimicking Annotator Annotation) in Toxic Comment Identification across Diverse Communities. arXiv preprint arXiv:2311.00203.

Theses

[T.2] **Senjuti Dutta**. (2024). Understanding Traits to Support Crowdworkers' Flexibility. PhD thesis, Department of Electrical Engineering and Computer Science, University of Tennessee, Knoxville, TN, USA. August, 2024.

[T.1] **Senjuti Dutta**. (2020). Enabling Efficient and Privacy-Preserving Task Matching For Cloud-Based Crowdsourcing. Master thesis, Department of Computer Science, University of Memphis, Memphis, TN, USA. July, 2020.

Professional Activity

- Lead Workshop Organizer, *The Future of Research in the Age of Human–AI Collective Intelligence*, **HCOMP & ACM Collective Intelligence 2026**.
  Conceived and initiated the workshop; developed its intellectual vision, thematic framework, program structure, invited-speaker program, poster session, and paper/poster selection process; and led the overall organization and coordination.
- Book Chapter Reviewer, **Design Patterns in Modeling and HCI, 2026**
  Reviewed Chapter 4 of *Design Patterns in Modeling and HCI*, a forthcoming book slated for publication by Oxford University Press.
- Workshop Co-Chair, **ACM Collective Intelligence and HCOMP 2026**
  Lead workshop planning, including writing the Call for Proposals (CFP), reviewing submissions, the whole workshop activities, reaching out to workshop organizers.
- Publicity and Networking Co-Chair, **HCOMP 2024**
  Led visibility and outreach initiatives to increase global participation and foster interdisciplinary connections.
- Workshop Organizer, **Deep Learning Indaba 2024**
  Co-led the planning and execution of a workshop named" Envisioning Humans in a Safe, AI Future" on inclusive and responsible AI; contributed writing all the content and some speaker coordination.
- Student Volunteer, **CSCW 2021, HCOMP 2021, CHI 2022**
  Supported operational logistics and attendee engagement at top-tier HCI conferences.
- Paper Reviewer, **CHI 2027, CHI 2026, HCOMP 2026, CHI 2025, CHI 2024, NAACL 2025, SIGIR 2025, HCOMP 2024, DIS 2024, WSDM 2025, CSCW 2023, ARR 2022**
  Provided detailed peer reviews for interdisciplinary conferences spanning HCI, AI, crowdsourcing, and communication systems.

Invited Talks

- *Interrelated Crowdworker Preferences: Implications for Flexible and Personalized Platform Design*, at the CHIRE (CHAI Human Information Retrieval) Seminar of **RMIT University**, Nov 12, 2024
- *Design Implications for Developing Flexible and Inclusive Crowdwork AI Systems*, at Participatory AI for Community Engagement (PACE 2024) Workshop of **HCOMP 2024**, Oct 19, 2024

Student Mentoring

- Md. Uzzal Talukder (August 2026- December 2026, PhD student at University of Tennessee, Knoxville)
- John Cheek (July 2026- August 2026, Undergraduate student at University of Tennessee, Knoxville)
- Emily Doherty (2025-2026, PhD student at University of Colorado, Boulder)

- Cara Spencer (2025-2026, PhD student at University of Colorado, Boulder)
- Amanda Hernandez Sandate (2025-2026, PhD student at University of Colorado, Boulder)
- Varun Naraganahalli Raghavendra (2025-2026, Masters student at University of Colorado, Boulder)
- Paula Akemi (2023-2024, PhD student at University of Toronto)
- Kelsey Stemmler (2023-2024, External research Assistant at the University of Toronto, currently a Msc candidate at University of Toronto)
- Conor O'Malley (2022-2023, Undergraduate student at University of Tennessee, currently a Patent Examiner at USPTO)

Media Coverage

- Featured in Postdoc Spotlight:
  Leading Conversations in Human-AI Intelligence
- Featured in the Dysoc newsletter:
  Dysoc welcoming postdoctoral fellow
- Featured in the University of Tennessee Dysoc newsletter:
  Spotlight on DySoC Members: Innovators Shaping the Future
- Featured in the University of Tennessee newsletter on March 29, 2022:
  Mobilizing Crowdwork: A Systematic Assessment of the Mobile Usability of HITs

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